

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE413: Computer Graphics

Batch: 18th

Semester: Spring 2023

Time: 1 Hour 30 Minutes

Date: 19/06/2023

Mid-Term Examination

Total Marks: 30

Answer any five questions of the followings.

5 × 6 = 30

1. (a) Explain the electrostatic deflection of the electron beam in a CRT. 3
(b) Describe the coordinate representation in computer graphics. 3
2. (a) Draw the architecture of a raster-graphics system with a display processor. 3
(b) Suppose an RGB raster system is to be designed using an 8-inch by 10-inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) do we need for the frame buffer? 3
3. (a) Differentiate between DDA algorithm and Bransenham's line drawing algorithm. 2
(b) Using the Bresenham's Line Drawing algorithm, generate and draw the coordinates of the pixel that lie on a line segment having the endpoints (20, 10) and (30, 18). 4
4. (a) Differentiate between Flood-fill algorithm and Boundary-fill algorithm. 2
(b) Consider a circle having the radius $r = 20$ and the circle centered on the coordinate origin. Draw the octets of the positive coordinates using Midpoint Circle algorithm. 4
5. (a) What are the importance of using Homogeneous Coordinates? 2
(b) Plot the region 1 of the ellipse centered at (0, 0) with radius $r_x = 12$ and $r_y = 10$ using Midpoint ellipse drawing algorithm. Assume that, the initial decision parameter in region 1 as $p_{1_0} = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$. 4
6. (a) Explain 2D rotation in computer graphics. Given a triangle with corner coordinates (0, 0), (6, 0) and (3, 4). Rotate the triangle by 90 degree anticlockwise direction and find out the new coordinates. 3
(b) Define X-shear in computer graphics. A Triangle with (2, 2), (0, 0) and (2, 0). Apply Shearing factor 3 on Y-axis and find out the new coordinates of the triangle. 3
7. (a) Define Viewport with example. Draw the 2D viewing – transformation pipeline. 3
(b) Let us assume, $X_{wmin} = 40, X_{wmax} = 100, Y_{wmin} = 50, Y_{wmax} = 100$ for window. $X_{vmin} = 50, X_{vmax} = 80, Y_{vmin} = 50, Y_{vmax} = 78$ for viewport. Now, consider a point $(X_w, Y_w) = (60, 70)$ on the window. Calculate that point on the viewport. 3

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 417: Compiler Design

Batch: 18th

Date: 21/06/2023

Mid-Term Examination

Semester: Spring 2023

Time: 1 hour 30 minutes

Total Marks: 30

Answer any three questions.

Marks: 10 x 3 = 30

1. (a) What is scanner and how it works? 02
(b) Write a program in flex to detect to detect an unsigned floating point number. 04
(c) What are the phases of compiler? Translate- the following statement into different phases: 04

position = value + rate x 75%.

2. (a) Divide the following program segment into appropriate lexemes 04
float limitedSquare(x)

```
{ float x  
    return (x <= -10.0 || x >= 10.0) ? 100 : x*x;  
}
```

- (b) Define and describe the terms: 06
RE, patterns, r+, r*, [a-zA-Z0-9], [az-] → 6

3. (a) What is lexical analyzer? Write the main task of lexical analyzer. 03
(b) What do you mean by ambiguity of grammar? Show that following grammar is ambiguous 03

S → if E then S |
if E then S else S

- (c) Write down various tokens, their informal description and sample lexemes. 04

4. (a) Differentiate between one-pass and multi-pass compiler. 03

- (b) Explain the following terms with proper example: 03
i) Associativity. of operators. ii) Precedence of operators. → 6

- (c) $expr \rightarrow expr - term \mid expr + term \mid term$ 04
 $term \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid \dots \mid 9$

Draw the annotated parse tree for the input $9 + 5 - 2$

5. (a) Consider an arithmetic expression (represented in infix notation) of integers and identifiers with the four binary operators +, -, *, /. Now construct an unambiguous grammar to evaluate the expression. Also verify your grammar 05 → 5

with an input string that contains all the operators mentioned above.

(b)

Define predictive parsing. Consider the following grammar:

type $\rightarrow$ simple | $\uparrow$ id | array[simple] of type

simple $\rightarrow$ integer | char | num dotdot num

input : array [num dotdot num] of integer Draw the predictive parsing table.

05

FACULTY OF BUSINESS ADMINISTRATION
BBA 321/MGT 417(CSE)/ MGT 419 (EEE): Operations Management/
Industrial Operational Management

24/06/2023

Mid-Term Examination

Semester: Spring 2023

Time: 1.5 Hours

Total Marks: 10x3=30

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answers to the questions should be neat and clean.
- Answer any three (3) of the following questions.

01. a. Define operations management and state the three basic functions of business management. 05
b. Explain the importance of learning about operations management. 05
02. a. Describe the key aspects of operations management decision making. 05
b. Identify the similarities and differences between production and service operation. 05
03. a. Explain the several ways that business organizations compete. 05
b. Determine the multifactor productivity for the combined input of labor and machine time using the following data: 05
- Output: 7040 units
Input:
Labor: \$1000
Materials: \$520
Overhead: \$2,000
04. // Determine the productivity for following cases: 05
I. Four workers installed 720 square yards of carpeting in eight hours.
II. A machine produced 70 pieces in two hours. However, two pieces were unusable.
b. Define strategy and tactics and state how to formulate strategy. 05

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 411: Artificial Intelligence

Batch: 18th

Semester: Spring 2023

Date: 05/07/2023

Mid Term Examination

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the following:

5x6= 30

- 1 (a) What is the significance of inference in the context of AI? 2
(b) How does AI contribute to game playing and the comprehension of natural language? 4
- 2 (a) How does the accumulation of knowledge contribute to the enhancement of intelligence? 2
(b) In which knowledge categories do the provided examples belong? 4x1 = 4
i. Knowing how to start the engine, shift gears, and operate the pedals.
ii. Knowing to trust a well-established brand when purchasing a product.
iii. Knowing that water boils at 100 degrees Celsius at sea level.
iv. Knowing how to balance, pedal, and steer.
- 3 (a) Find the meaning of the statement, $S = (W \vee X) \wedge (W \vee \neg X) \vee W$ if W is True and X is False. 2 7
(b) Let's consider a perceptron with two inputs, x_1 and x_2 , and corresponding weights $w_1 = 0.5$, $w_2 = 0.2$, threshold = 0.2 and learning rate = 0.1. Implement this perceptron to solve AND gate. 4 $\wedge$
 $\vee$
 $\rightarrow$
 $\leftrightarrow$
- 4 (a) Create a semantic network to represent the following information: 2
• Apples are fruits. .
• All fruits grow on trees.
• Oranges are fruits.
• Oranges are orange in color.
• Oranges have Vitamin C.
(b) Write the predicate logic for the following statements: 4x1 = 4
i. Some students love mathematics.
ii. There exists a person who speaks multiple languages.
iii. For every student, there is a teacher who inspires them.
iv. There exists a city where every citizen speaks at least two languages.
- 5 (a) Write about step and sigmoid activation function. 2
(b) Suppose a Hopfield network is trained with the input vector (1 1 -1 -1) 4x1 = 4
for auto-association. Find the following:
i. Calculate the weight matrix for the Hopfield network.
ii. Test the network with the input vector $x = (1 \ 1 \ -1 \ -1)$.
Determine the output obtained after one iteration.

- iii. Test the network with an input vector that has one missing component, for example, $x = (1 \ 1 \ -1 \ ?)$. Determine the output obtained after one iteration.
- iv. (d) Test the network with an input vector that has two mistaken components, for example, $x = (1 \ -1 \ -1 \ -1)$. Determine the output obtained after one iteration.

- 6
- (a) Define frame. 1
 - (b) Draw the diagram of an associative memory network. 2
 - (c) Draw a frame for the following expression: 3
 "There is a specific breed of dog known as a Labrador Retriever, which has a thick double coat, is known for its friendly temperament, and is often used as a guide dog."
- 7
- (a) Outline the weaknesses or shortcomings of Propositional Logic (PL). 2
 - (b) How do biological neural networks and artificial neural networks relate to each other? 2
 - (c) Which rules of inference can be used to derive the conclusions in these examples? 0.5x4 = 2
 - i. Premise 1: If it is raining, then the ground is wet.
 Premise 2: The ground is not wet.
 Conclusion: Therefore, it is not raining.
 - ii. Premise 1: If it is sunny, then I will go for a walk.
 Premise 2: If I go for a walk, then I will feel refreshed.
 Conclusion: Therefore, if it is sunny, then I will feel refreshed.
 - iii. Premise: The grass is green.
 Conclusion: Therefore, the grass is green or the sky is blue.
 - iv. Premise 1: A is true
 Premise 2: B is true
 Conclusion: Therefore, A and B are true.



Faculty of Science, Engineering and Technology Department of Computer Science and Engineering

CSE 415: Communication Engineering

Batch : 18th

Date: 08/07/2023

Mid-Term Examination

Semester: Spring 2023

Time: 1 Hour 30 Minutes

Total Marks : 30

Answer any five questions:

1. (a) Describe the services of the Data-link layer.
(b) Draw and explain an address resolution protocol packet.
2. (a) Explain all types of addressing used in Data-link layer.
(b) Illustrate the processes of error detection in block coding.
3. (a) What is redundancy? Narrate all types of errors.
(b) Distinguish between error detection and error correction.
4. (a) Draw and explain encoder and decoder for simple parity-check code.
(b) Describe the states of FSM for stop-and-wait protocol.
5. (a) Explain all the fields of HDLC frames.
(b) Draw a flow chart to calculate traditional checksum.
6. (a) Briefly explain Password Authentication Protocol.
(b) Using the following specifications, draw a finite state machine with three states (I, II, and III), five events, and six actions:
 - I. If the machine is in state I, two events can occur. If event 1 occurs, the machine moves to state II. If event 2 occurs, the machine performs actions 1 and 2 and moves to state III.
 - II. If the machine is in state II, two events can occur. If event 3 occurs, the machine remains in state II. If event 4 occurs, the machine moves to state III.
 - III. If the machine is in state III, three events can occur. If event 2 occurs, the machine remains in state III. If event 3 occurs, the machine performs actions 1, 2, 4, and 5 moves to state II. If event 5 occurs, the machine performs actions 1, 2, and 6 and moves to state I.
7. (a) Draw the taxonomy of multiple-access protocols. Narrate random access protocol.
(b) There are only three active stations in a slotted Aloha network: A, B, and C. Each station generates a frame in a time slot with the corresponding probabilities $p_A = 0.2$, $p_B = 0.3$, and $p_C = 0.4$ respectively.
 - I. What is the throughput of each station?
 - II. What is the throughput of the network?

S-C+

Answer any five of the following questions as per Alphabetical order.

5 × 6 = 30

1. (a) How many ways Cascading Style Sheet (CSS) can be inserted in a Hyper Text Mark-up Language (HTML) document? 2
(b) Consider there are five fields in a navigation-bar such as- Home, About, Education, Hobby and Contact. Write the Hyper Text Mark-up Language (HTML) code for this navigation-bar. Must use Hover attribute in this navigation-bar and use the Cascading Style Sheet (CSS) and JavaScript. 4
2. (a) Briefly explain the User input, Interaction and Portals web applications with appropriate examples. 3
(b) Consider a div tag with id div_1. Write the following statements using CSS: 3
 - i. Give the div top margin size 2.5 pixel and margin position in left
 - ii. Use padding in left side is 15 pixel and right side is 10 pixel
 - iii. Add background image named "my_img.jpg"
3. (a) List the features of JavaScript. "JavaScript is inherited from Java programming language." (True/False). 2
(b) Consider an HTML form that has two input fields and two buttons. These fields take two numbers from users. Two buttons are named "MULT" and "DIV". If anyone clicks the button "MULT", it shows the multiplication of two numbers into an alert box and if they click on the "DIV" button, it shows the division of two numbers without using an alert box. 4
4. (a) Write the main purposes of an Agile process model. 2
(b) Suppose a multinational company known as "ABC Inc." exists. This company has several disciplines such as: Admin, Accounts, Registrar, Staff, Operations, ICT, etc. So, their requirements can change in a very short period of time. For this reason, they want to develop an "Enterprise Resource Planning (ERP)" system. 4
Now, to meet these requirements, you need to choose the appropriate model of web application between the Waterfall Process Model and the Incremental Process Model. You need to defend your choice and explain the mechanisms of your choice.
5. (a) Consider a software company X wants to develop a web application named 'Personal Bio' for Y customer. Explain the process model which is suitable for this system. 3
(b) Write the HTML code for the following list: 3

- List 1
 - i. Sub list 1.1
 - ii. Sub list 1.2
- List 2
 - A. Sub list 2.1
 - B. Sub list 2.2
 - i. Sub list 2.2.1
- List 3
 - Sub list 3.1
 - Sub list 3.2

6 (a) Write the best practice approaches of web applications. 3

~~(b)~~ Draw the basic structure of HTML format with example. 3

7 (a) What are the methods of web engineering? 2

~~(b)~~ Briefly explain the attributes of web applications. 4